4.3. $T_0 = mg + m\frac{v_0^2}{l}$, $I = mv_0$ 解得 $I = \sqrt{0.735} \approx 0.857 N\cdot s$

4.10 设阻力为 f.

(1) $-f\cdot\frac{L}{2} = 0 - \frac{1}{2}mv_0^2$, $-fL = 0 - \frac{1}{2}mv_f^2$ 得 $v_f = \sqrt{2}v_0$.

(2) 动量守恒, $mv_0 = (m+M)v$ … ①

~~动能定理~~, 功能关系, $-fx = \frac{1}{2}(m+M)v^2 - \frac{1}{2}mv_0^2$. … ②

解得 $x = \frac{M}{2(m+M)}L$.

(3) 由动能定理, $fs = \frac{1}{2}Mv^2 - 0$, 而由(1)得 $f = \frac{mv_0^2}{L}$

∴ $s = \frac{mM}{2(m+M)^2}L$.

4.17. $v_0 = 500\ m/s$ 很大, 忽略极短时间内摩擦力冲量, 射入到射出目穿间可视为动量守恒

$mv_0 = mv + Mv_1$ 解得 $v_1 = 0.8 m/s$.

$\mu Mgs = \frac{1}{2}Mv_1^2$ 解得 $\mu = 0.16$.

(2) $\Delta E_k = \frac{1}{2}m(v_0^2 - v^2) = 240J$.

4.19 (1) 由水平方向动量守恒, $mx_1 = Mx_2$ $L = \frac{h}{\tan\alpha}$ x_1, x_2 均为相对地面位移.

由几何关系 $x_1 + x_2 = L$

解得 $x_2 = \frac{m}{M+m}h\frac{1}{\tan\alpha}$

(2) 设小车 v_2 向右, 滑块 v_r 沿斜面向左下:

$mv_{rx} = Mv_2$

$\tan\alpha = \frac{v_{ry}}{v_{rx}+v_2}$ 故 $v_{ry} = \frac{M+m}{m}v_2\tan\alpha$

$mgh = \frac{1}{2}Mv_2^2 + \frac{1}{2}m(v_{rx}^2 + v_{ry}^2)$ 解得 $v_2 = \sqrt{\frac{2m gh}{M(M+m) + (M+m)^2\tan^2\alpha}} \cdot m$